



ENVIRONMENTAL SCIENCE

Four billion people lack safe water

Water safety is a key challenge compounded by climate risks and data gaps

By **Rob Hope**

In 2020, more than 2 billion people were estimated to lack access to safely managed drinking water services (1). However, the existing data on safely managed drinking water services are scarce especially in low- and middle-income countries. On page 784 of this issue, Greenwood *et al.* (2) report that an estimated 4.4 billion people lack safe drinking water across 135 low- and middle- income countries, which is more than double the global estimate made in 2020. Their approach involves geospatial modeling of Earth observation—information from land-based, airborne, and satellite data—and household surveys. The authors identify primary factors, including fecal con-

tamination, that affect drinking water safety, which can support policy-makers to improve delivery of safe drinking water services.

Safely managed drinking water services must satisfy three criteria: They should be available on demand, accessible on premises, and free of priority contaminants such as *Escherichia coli* and specific chemicals (1). Established policies and practices often focus on improving access through building new water infrastructures. However, water service is more than just providing access. According to the Human Rights to Water and Sanitation resolution declared by the United Nations, water services must ensure sufficient quantity, safety, reliability, physical proximity, affordability, and nondiscrimination (3). These goals are challenging in rural areas of Africa

and Asia and in sparsely populated regions where safe drinking water services on premises are costly and complicated to maintain.

Greenwood *et al.* mapped the contributions of the three safe water criteria to assess the role of climate variability, environmental change, and human activities. The results identify fecal contamination as the primary limiting factor in achieving safely managed drinking water for almost half of the population studied. However, insufficient data were available to model contaminations from chemical sources, such as arsenic and fluoride, which limits discussion of potential pollution from industry or agriculture (4).

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A woman refills drinking water jars from the water spout in Lalitpur, Nepal. Women are one of the groups that are exposed to water insecurity.

treat, and deliver water from distant sources to cities because local resources have been overexploited or polluted. Competition for water under increasing climate variability and urban population growth has contributed to annual transfers of around 16 billion m³ of water from rural areas to urban centers worldwide (7).

Thus, improving the quality of data-monitoring systems is required to support policy and practice to provide reliable and affordable drinking water services. Relying on data collected from national household surveys conducted only every 5 to 10 years is a common practice. However, such household survey data are only snapshots and miss impacts from climate shocks or policy change. For example, in August 2019, India announced a national program to provide functional household tap connections to 1.4 billion people. By 2024, over 77% of households reported a tap connection, which is a major increase from 16% in 2019 (8).

Beyond the analysis of Greenwood *et al.*, further consideration of safely managed drinking water systems in public facilities is needed. Household data do not reflect an individual's use of drinking water in public domains. For example, individuals in the same household may use different drinking water supplies if they spend part of their day in a school, hospital, or clinic. Despite attempts to consolidate data for public facilities (9), there remains a major global data gap. Predictions based on household data alone could have unanticipated effects on policy if the collected data are incomplete, time-bound, or missing information about an individual's use of public water services.

To overcome the limits of current global monitoring of drinking water in homes or public facilities, new data systems have emerged from urban and rural water service providers. The World Bank's International Benchmarking Network for Water and Sanitation Utilities (IBNET) is a data source for over 5000 urban utilities in 150 countries worldwide. In Latin America, the Rural Water and Sanitation Information System (SIASAR) operates in multiple countries in both rural and urban areas to provide a database on water supply and sanitation services as an information tool. Uptime is a global program that guarantees drinking water to 5 million rural people in 16 countries. The program provides results-based funding for water service providers who meet three quarterly performance metrics—reliability, water volume, and local revenue (10). These platforms can provide longitudinal data on the observed delivery

of safely managed drinking water. Future global modeling studies may consider using these data to capture the effects of climate shocks and policy changes on water services. More accurate modeling can bring a robust monitoring system to inform global and local policy-makers.

In response to this daunting global challenge, subsequent actions and investments need to consider three factors: the right scale, financial sustainability, and water inequalities. Global data provide a general view, whereas national mandates and responsibilities are critical for delivering, monitoring, and maintaining safe water services. More robust, timely, and accurate national data systems are needed to improve water services. Past financial investments have failed to deliver desired outcomes, and future projections are not affordable for middle- and low-income countries (11). New ideas are needed to shift from financing infrastructure to funding maintenance of safe water services. In addition, a 4 billion target must reflect the social inequalities of water services that disproportionately fall on women, girls, pastoralists, and other groups who often live in water-insecure environments (12). Policies must balance between serving the greatest number cost-effectively and ensuring equity, so that the most vulnerable are neither excluded nor benefit last. ■

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Additionally, Greenwood *et al.* show that environmental factors such as high annual average temperatures and seasonality of precipitation negatively affect safely managed drinking water. Further, land use, vegetation, and bedrock, which influence the storage and movement of groundwater, affect overall water resource availability. These observations attempt to understand the local effects of the hydrological cycle, environmental conditions, and human activities on water security (5). For example, longitudinal modeling—surveying the same household over time—in Africa has shown that seasonal rainfall patterns, such as dry spells in wet seasons or wet spells in dry seasons, influence water demand. This then affects seasonal revenue—generated from the tariffs paid by users—and thus the financial sustainability of maintaining drinking water services (6).

Underground stores of water such as aquifers and surface water bodies like rivers are faced with unprecedented threats in all regions owing to human and climate influences. This has increased costs to pump,